

VERONA VILLAFRANCA, Italy

WMO: 160900

Lat: 45.396N

Lon: 10.889E

Elev: 239

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.1	26.2	6.7	7.8	41.2	13.8	11.2	39.5	20.5	44.7	17.8	43.1	3.2	0	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	93.3	72.7	91.1	72.5	88.2	71.4	78.4	84.3	76.6	83.5	75.0	82.6	5.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	141.9	80.5	75.0	132.4	79.0	73.1	124.0	77.9	42.0	85.3	40.2	84.7	38.6	82.9	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	15.6	12.7	DB	19.0	96.0	3.8	3.0	16.3	98.2	14.1	99.9	12.0	101.6	9.2	103.8	(4)
(5)			WB	18.2	81.3	3.7	2.7	15.5	83.2	13.3	84.8	11.2	86.4	8.5	88.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	37.8	41.2	48.8	56.2	65.0	73.0	76.9	76.0	67.6	57.9	47.8	39.0		
	DBStd	14.84	4.92	5.21	5.68	5.57	5.47	5.80	4.68	5.03	5.26	5.63	6.35	5.86		
	HDD50	1188	378	248	87	12	0	0	0	0	0	8	112	343		
	HDD65	3934	842	666	501	268	67	8	0	1	33	226	515	807		
	CDD50	3875	0	2	51	197	466	689	833	807	527	254	47	2		
	CDD65	1146	0	0	0	3	68	247	368	343	110	7	0	0		
	CDH74	11257	0	0	4	58	598	2406	3910	3458	796	27	0	0		
	CDH80	4346	0	0	0	4	136	897	1660	1463	184	2	0	0		
	WSAvg	4.6	3.7	4.6	5.4	5.7	5.4	5.0	4.8	4.5	4.4	4.0	4.2	3.6		
	PrecAvg	30.3	1.7	1.7	2.1	2.6	3.0	3.2	2.4	3.4	2.8	3.1	2.9	2.0		
(15) Precipitation	PrecMax	43.1	6.9	6.6	5.9	7.0	8.0	7.1	5.6	15.3	7.3	9.3	7.4	5.5		
	PrecMin	11.5	0.0	0.0	0.0	0.5	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	6.3	1.5	1.5	1.5	1.2	1.8	1.7	1.4	2.8	1.9	2.3	1.5	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	61.1	71.9	79.1	88.1	94.0	96.8	97.1	88.0	77.1	65.4	56.6		
		MCWB	48.5	50.3	53.8	60.6	66.7	72.8	71.9	73.0	69.6	62.9	58.3	49.4		
	2%	DB	50.4	56.7	66.5	75.3	84.0	91.4	93.3	93.4	84.8	73.5	62.1	52.2		
		MCWB	44.8	47.5	53.4	58.8	65.5	72.1	73.1	72.4	68.2	63.7	57.0	47.5		
	5%	DB	48.2	53.5	62.9	71.7	80.4	88.1	90.9	91.0	82.0	70.2	59.3	50.0		
		MCWB	43.7	45.7	52.0	57.1	63.6	70.6	73.0	72.5	66.8	61.6	55.8	46.3		
	10%	DB	46.1	50.2	59.5	68.2	77.0	85.7	88.0	87.8	78.7	67.7	57.2	47.3		
		MCWB	42.6	44.6	50.2	56.2	62.7	69.5	71.7	71.8	65.3	60.5	54.2	44.7		
	0.4%	WB	51.1	53.0	58.1	63.7	72.6	79.0	81.0	81.2	74.7	69.2	62.7	53.3		
		MCDB	53.2	59.3	65.3	72.8	78.9	84.2	86.1	86.6	79.2	72.1	63.4	54.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.0	49.8	55.6	61.0	69.3	76.3	78.4	78.5	72.4	66.2	58.9	49.5		
		MCDB	48.6	53.5	62.5	69.7	77.0	83.5	84.8	84.0	78.2	69.4	60.4	50.7		
	5%	WB	44.9	47.8	53.8	59.4	67.1	74.4	76.6	76.6	70.3	64.2	56.8	46.9		
		MCDB	46.5	50.9	60.4	67.8	74.7	82.6	83.7	83.5	77.1	67.9	58.8	48.3		
	10%	WB	42.9	46.0	52.0	57.9	65.2	72.6	74.9	74.9	68.8	62.4	54.8	45.3		
		MCDB	44.5	48.8	57.6	65.5	72.9	81.1	82.9	82.2	75.0	65.8	56.9	46.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	15.4	17.7	18.2	18.4	18.5	19.3	19.3	18.2	15.2	12.5	12.5		
		MCDBR	16.7	21.1	23.8	24.4	23.2	21.2	21.6	22.4	21.6	18.7	14.2	16.0		
		MCWBR	11.8	13.0	12.8	10.6	9.2	9.1	9.3	9.9	9.1	10.2	10.1	11.5		
		MCDBR	13.1	16.7	20.1	20.6	20.3	19.7	19.5	20.1	18.5	15.1	10.9	11.5		
	5% WB	MCWBR	10.4	11.5	12.0	10.5	9.8	9.6	9.8	10.5	9.9	9.8	8.8	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.348	0.377	0.422	0.457	0.456	0.481	0.464	0.453	0.441	0.430	0.380	0.342		
	taud		2.377	2.271	2.150	2.086	2.138	2.116	2.170	2.216	2.218	2.202	2.307	2.405		
	Ebn at Noon		236	249	253	255	259	253	255	253	245	226	219	225		
	Edh at Noon		24	32	42	48	47	49	46	42	39	34	26	22		
(44) All-Sky Solar Radiation	RadAvg		449	706	1119	1429	1768	2020	2072	1769	1306	792	468	386		
	RadStd		71	105	152	156	165	109	112	117	87	112	68	50		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	N/A	N/A	+1.67	N/A	N/A	N/A	-200	+292	+184
(47)	Regional (49 neighbors)	+0.61	N/A	N/A	+1.01	N/A	N/A	N/A	-167	+117	+18

Nomenclature: See separate page